

CLOUD + DEVOPS + INFRASTRUCTURE-AS-GOVERNANCE ENGINEER

Official Student Workbook

AWS • DevOps • Terraform Governance Engineering

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PROGRAM IDENTITY

Program Name

Cloud + DevOps + Infrastructure-as-Governance Engineer

Student

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Training Model

Build → Automate → Govern

Delivery Structure

- Friday — Core Engineering Concepts
 - Sunday — Practical Infrastructure Labs
 - Flexible Session — Terraform + Governance Automation
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PROGRAM PHILOSOPHY

Most training programs teach:

- AWS
- DevOps tools
- Infrastructure deployment

This program goes further.

This roadmap trains students to think like:

- ✓ Cloud Architects
- ✓ DevOps Engineers
- ✓ Infrastructure Automation Engineers
- ✓ Governance-Focused Operators

The goal is not only to deploy infrastructure.

The goal is to:

- Control infrastructure
- Govern infrastructure
- Standardize deployments
- Automate operations
- Reduce risk
- Improve visibility
- Build scalable systems

CORE ENGINEERING PRINCIPLE

“If you can’t code it, you don’t control it.”

This workbook follows a strict engineering pattern:

Stage 1

Build infrastructure manually

Stage 2

Understand architecture deeply

Stage 3

Automate infrastructure using Terraform

Stage 4

Apply governance and operational control

PHASE 1 — ARCHITECTURE + TERRAFORM FOUNDATION

Month 1

Goal:

Everything built manually → then rebuilt using Terraform

WEEK 1 — VPC + TERRAFORM INTRODUCTION

Friday Session

Architecture Fundamentals + Terraform Introduction

Core Learning Objectives

Students must understand:

- What a VPC is
 - CIDR block planning
 - Public vs private subnets
 - Internet Gateway purpose
 - Route tables
 - Terraform fundamentals
 - Infrastructure-as-Code philosophy
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Architecture Concepts

What is a VPC?

A Virtual Private Cloud (VPC) is an isolated network inside AWS.

It allows organizations to:

- Control networking
 - Segment workloads
 - Secure applications
 - Design scalable architectures
-

CIDR Example

Network	CIDR
VPC	10.0.0.0/16
Public Subnet	10.0.1.0/24
Private Subnet	10.0.2.0/24

Friday Practical Lab

Task 1 — Install Terraform

Students Must:

- Install Terraform
- Verify installation
- Configure AWS CLI credentials

Verification Command

```
terraform version
```

Task 2 — Create First Terraform File

File Name

```
main.tf
```

Basic Terraform Provider Example

```
provider "aws" {  
    region = "us-east-1"  
}
```

Sunday Session

Manual VPC Deployment

Students Must Build:

- ✓ VPC
- ✓ Public subnet
- ✓ Private subnet
- ✓ Internet Gateway
- ✓ Route Tables

AWS Console Tasks

Task	Status
Create VPC	☐
Create Public Subnet	☐
Create Private Subnet	☐
Attach Internet Gateway	☐
Configure Route Table	☐

Flexible Session

Terraform VPC Lab

🎯 Objective:

Rebuild the entire VPC using Terraform

Terraform Tasks

Students Must:

- Define VPC resources
 - Create subnets
 - Attach IGW
 - Configure route tables
 - Run terraform init
 - Run terraform plan
 - Run terraform apply
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Reflection Questions

1. Why is Infrastructure-as-Code important?

2. What risks exist in manual infrastructure deployment?
3. Why should networking be standardized?
4. What operational advantages does Terraform provide?

WEEK 2 — EC2 + AUTOMATION

Friday Session

EC2 + Storage Fundamentals

Core Concepts

EC2

Elastic Compute Cloud provides virtual servers inside AWS.

EBS

Elastic Block Store provides persistent storage volumes.

Friday Lab

Students Must:

- ✓ Launch EC2 manually
 - ✓ Configure security groups
 - ✓ Attach EBS volume
 - ✓ Connect using SSH
-

Sunday Session

Application Deployment

NGINX Deployment Lab

Students Must:

- Install NGINX
 - Configure web server
 - Deploy test web page
 - Verify browser access
-

Linux Commands

```
sudo yum update -y
sudo yum install nginx -y
sudo systemctl start nginx
sudo systemctl enable nginx
```

Flexible Session

Terraform EC2 Automation

Terraform Objectives

Students Must:

- ✓ Deploy EC2 via Terraform
 - ✓ Automate provisioning
 - ✓ Configure variables
 - ✓ Create reusable infrastructure
-

Governance Insight

“Manual deployment creates inconsistency.

Automation creates operational reliability.”

WEEK 3 — LOAD BALANCING + TERRAFORM STATE

Friday Session

Application Load Balancer (ALB)

Concepts

ALB Purpose

Distributes traffic across multiple instances.

Benefits

- ✓ High Availability
 - ✓ Fault Tolerance
 - ✓ Scalability
-

Sunday Session

Multi-Instance Architecture

Students Must:

- Deploy multiple EC2 instances
 - Connect instances to ALB
 - Test load balancing
-

Flexible Session

Terraform State Governance

Terraform State Topics

Students Learn:

- What state files are
 - Why state matters
 - Remote state management
 - S3 backend configuration
 - State versioning
-

Governance Principle

“State = Infrastructure Truth”

If state becomes corrupted:

- Infrastructure drift occurs
 - Deployments become dangerous
 - Governance weakens
-

WEEK 4 — IAM + TAGGING GOVERNANCE

Friday Session

IAM Fundamentals

Core Topics

- ✓ IAM Users
 - ✓ IAM Roles
 - ✓ IAM Policies
 - ✓ Least Privilege
 - ✓ MFA
-

Sunday Session

Architecture Hardening

Security Tasks

Students Must:

- Audit security groups
 - Enable VPC Flow Logs
 - Review inbound/outbound traffic
 - Identify security risks
-

Flexible Session

Terraform Governance Engineering

Governance Automation Tasks

Students Must:

- ✓ Create IAM via Terraform
 - ✓ Enforce tagging standards
 - ✓ Apply governance rules
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Mandatory Tags

Tag	Purpose
Owner	Accountability
Environment	Environment identification
CostCenter	Financial tracking

FinOps Principle

“No tags = No visibility = No control”

Without tagging:

- Costs become difficult to track
- Ownership becomes unclear
- Governance weakens

PHASE 2 — DEVOPS + TERRAFORM INTEGRATION

Month 2

🎯 Goal:

Infrastructure + Code + Pipelines working together

Weekly Structure

Day	Focus
Friday	DevOps Concepts
Sunday	Pipeline Build
	Flexible Terraform Integration

CI/CD Concepts

Students learn:

- Continuous Integration
- Continuous Deployment
- GitHub Actions
- Automated workflows
- Infrastructure pipelines

Terraform Pipeline Flow

```
Developer Push
  ↓
GitHub Actions
  ↓
terraform plan
  ↓
Review
  ↓
terraform apply
```

Governance Principle

“No manual deployments allowed.”

Why?

Because uncontrolled deployments:

- Increase risk
 - Create inconsistency
 - Reduce auditability
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PHASE 3 — PLATFORM ENGINEERING

Month 3

🎯 Goal:

Standardized reusable infrastructure

Terraform Modules

Students build:

- ✓ VPC modules
 - ✓ EC2 modules
 - ✓ Reusable architecture templates
-

Multi-Environment Deployment

Students Deploy:

- Development
 - Staging
 - Production
-

Engineering Principle

“Standardization = Scalability”

PHASE 4 — MONITORING + GOVERNANCE

Month 4

🎯 Goal:

Operational visibility and control

Monitoring Stack

Students Build:

- ✓ CloudWatch Dashboards
 - ✓ Alerts
 - ✓ SNS Notifications
 - ✓ Operational Monitoring
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Governance Monitoring

Focus Areas

- Tag enforcement
- Access control
- Cost-aware monitoring
- Infrastructure visibility

Critical Operational Principle

“If you can’t see it, you can’t manage it.”

FINAL ENGINEERING OUTPUT

After 4 months, students will have built:

- ✓ Full AWS architecture
 - ✓ Terraform-managed infrastructure
 - ✓ CI/CD automation pipelines
 - ✓ Governance-enforced deployments
 - ✓ Monitoring systems
 - ✓ Multi-environment infrastructure
 - ✓ Cost visibility systems
 - ✓ Secure IAM architecture
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FINAL PROFESSIONAL TRANSFORMATION

This roadmap is NOT standard cloud training.

This is:

Infrastructure-as-Governance Engineering

Students are no longer only learning:

- AWS
- DevOps
- Terraform

They are learning:

- Governance Engineering
- Operational Control
- Infrastructure Standardization
- Enterprise Automation

- Platform Reliability

FINAL MESSAGE TO STUDENTS

Cloud engineering is no longer only about deployment.

Modern engineers must:

- Automate infrastructure
- Govern environments
- Control risk
- Standardize systems
- Build operational visibility

This program was designed to develop engineers who can operate at enterprise level.

Build carefully.

Automate intelligently.

Govern consistently.

Mthulisi Zulu